

AareML

Early Warning System for Swiss River Fisheries

Canton Zurich Innovation Sandbox — Phase III

T.N. & O.N. — AI & ML

Thiago Nascimento (Eawag) · Patrick Hager (Fischerei) · [Sensor expert]

The Problem

“Every day during heat season we manually measure water temperature — and decide whether to launch a full fish rescue operation within 24 hours.”

— Patrick Hager, President, oldest Swiss fishery

- 1 Rescue decisions made manually, daily
- 2 15-minute window to relocate fish
- 3 Known critical sites, zero automated alerts

Today: daily manual checks → 24h rescue decision · Target: 7 days automated

7-day temperature early warning for Swiss river fisheries

Predict

AI model forecasts water temperature 7 days ahead at critical river locations

Alert

Mobile app notifies fisheries and AWEL when threshold is approaching

Act

Fish relocate within 15-minute window before conditions turn critical

Prototype: AareML — 14-day DO + temperature prediction validated on 86 Swiss gauges. Retraining for temperature focus and 7-day horizon.

We have the model. We need the data.

What we have

- ✓ Working AI model (AareML, validated on CAMELS-CH-Chem)
- ✓ Scientific partner — Eawag (NAWA FRACHT data)
- ✓ Domain partner — Patrick's fishery (operational validation)
- ✓ Novel sensor partner — floating sensor prototype
- ✓ Transfer learning: deploy at ungauged sites with only metadata

What the sandbox unlocks

- Structured data access agreement with AWEL (150 cantonal monitoring sites)
- Real-time AWEL temperature feeds (currently internal only)
- Regulatory pathway: ML forecast → official fisheries advisory
- PFAS monitoring integration (Phase 2)

Four complementary partners

T.N. & O.N.

AI & Machine Learning

AareML prototype, 18 notebooks, UBELIX HPC

Thiago Nascimento

Eawag

CAMELS-CH-Chem dataset co-author, NAWA
FRACHT access, scientific validation

Patrick Hager

Fischerei

Domain expertise, go/no-go rescue ops, oldest
Swiss fishery

[Sensor Expert]

Novel Sensor

New floating depth-aware temperature sensor
prototype for critical river points

Three specific asks

1

Data MOU

Structured access to AWEL real-time temperature feeds

2

Regulatory pathway

Framework for ML-based fisheries advisories

3

PFAS pilot

Early integration of emerging contaminant monitoring

Outcome: operational 7-day early warning system for Canton Zurich fisheries — first in Switzerland